Facilities Study Report

For

Physical Interconnection of

PJM Generation Interconnection Request Project ID AF2-143

"POWERTON-NEVADA 345 KV"

Revision 2: December 2024

Introduction

This Facilities Study has been prepared in accordance with the PJM Open Access Transmission Tariff Part VII, and, if applicable, the Application and Studies Agreement between the Project Developer and PJM Interconnection, LLC (PJM or Transmission Provider (TP)). The Transmission Owner (TO) is ComEd.

A. Transmission Owner Facilities Study Summary

1. PROJECT DESCRIPTION

The Project Developer has proposed a solar Generating Facility located in La Salle, Illinois with a designated PJM Project ID of AF2-143.

This project will share Interconnection Facilities with Osagrove Flats 150 MW Wind Farm project (PJM New Service Request Project AD2-038) and will share the same Point of Interconnection.

The AF2-143 project is a 150 MW (MFO) Generating Facility with 90 MW recognized by PJM as Capacity.

2. POINT OF INTERCONNECTION (POI)

The Generating facility will interconnect with the ComEd transmission system via a direct connection into the TSS 918 DANA 345 kV substation.

The proposed generation interconnection is shown on the single line diagram in Attachment #1.

AF2-143 will share interconnection facilities with AD2-038. The combined output of all Generating Facilities at the Point of Interconnection is 300 MW.

Queues	MFO
AD2-038	150 MW
AF2-143	150 MW
Total Output	300 MW

3. POINT OF CHANGE IN OWNERSHIP

The Point of Change in Ownership will be located at the first dead end structure inside the TSS 918 Dana Substation fence line.

4. SCOPE OF PROJECT DEVELOPER INTERCONNECTION FACILITIES

Project Developer will design, build, own, operate and maintain the Project Developer Interconnection Facilities on Project Developer's side of the Point of Change in Ownership (PCO). This includes, but is not limited to:

At the planned 345 kV Osagrove Flats substation, in general, the following are project specific notes:

- New 345 kV AF2-143 collector substation facilities shall comply with all applicable portions of PJM Protection Standards (PJM Manual 7) and PJM Transmission and Substation Design Subcommittee Technical Requirements.
- New 345 kV L91801 generator lead from the generating facility 345 kV side to the existing L91801.
- Metering is required to be installed per ComEd & PJM standards per Section 7 of this study.
- Project Developer shall provide limiting Transmission Facility ratings for their portion of 345 kV L91801, in accordance with NERC FAC-008, FERC Order 881 and PJM Operational requirements for normal and emergency ratings from -55F to 130F in 5F increments.
- All Developer Interconnection Facilities shall comply with applicable requirements of Exelon Utilities Transmission Facility Interconnection Requirements dated January 1, 2024 and ComEd Interconnection Guidelines (For Generators at Transmission Level) dated December 16, 2021. The ComEd Interconnection Guidelines (For Generators at Transmission Level) dated December 16, 2021 and Exelon Utilities Transmission Facility Interconnection Requirements dated January 1, 2024 are available on the PJM website.
- One (1) Main Power Transformer (MPT-2), a three-winding transformer with a 345 kV (Hi-Y Grd) / 34.5 kV (Low-Y Grd)/ 13.8 kV (Tert-Delta). Z ~ 9.0% on 99 MVA base. 99/132/165 MVA ONAN/ONAF/OFAF, X/R = 40.
- One (1) new Project Developer Owned 345 kV gas circuit breaker.
- Project Developer to provide to ComEd test reports for the Project Developer owned transformers, including the equipment's %Z impedance and load loss, and the impedance and length of the generator lead line from the generating facility to PCO.
- Project Developer shall provide shunt reactive compensation as required by the PJM Interconnection studies.
- At AF2-143 collector station, in general, Project Developer to follow section 6.1 (Design F) of the latest version of ComEd Interconnection Guidelines for Generators at Transmission Level (Rev 2: Effective 12/16/2021).
- New gas circuit breakers to auto trip and isolate for critical gas level.
- ComEd Protection and Control Engineering must review all Project Developer relay protection design drawings and relay settings.
- Project Developer equipment impedance and/or test data must be provided to ComEd Protection and Control Engineering to model in a short circuit program. Examples include lines, transformers (include percent impedance and load loss), solar panels, and inverters.
- Modify 345 kV L91801 relay protection schemes to accommodate new interconnection AF2-143. Line terminal relay types to be the same as ComEd terminal relays. This includes relay firmware versions.

- The Generating Facility shall be designed to remain in service (not trip) for voltages and times as specified for the Eastern Interconnection in Attachment 1 of NERC Reliability Standard PRC-024-1, and successor Reliability Standards, for both high and low voltage conditions, irrespective of generator size, subject to the permissive trip exceptions established in PRC-024-1 (and successor Reliability Standards).
- The Generating Facility shall be designed to remain in service (not trip) for frequencies and times as specified in Attachment 2 of NERC Reliability Standard PRC-024-1, and successor Reliability Standards, for both high and low frequency condition, irrespective of generator size, subject to the permissive trip exceptions established in PRC-024-1 (and successor Reliability Standards).
- The solar Generating facility shall provide SCADA capability to transmit data and receive instructions from the Transmission Provider to protect system reliability. The Transmission Provider and the solar generation facility Project Developer shall determine what SCADA information is essential for the proposed solar generation facility, taking into account the size of the facility and its characteristics, location, and importance in maintaining generation resource adequacy and transmission system reliability in its area.
- Project Developer to include over/under frequency and voltage protection at solar farm collector bus.
- Suggested settings will be provided by ComEd. Under-frequency settings are to comply with MAIN Guide 1B.
- Dual bus protection for 34.5 kV bus.
- Relay and protective equipment, telecommunications equipment, and Supervisory Control and Data Acquisition (SCADA) to comply with the ComEd Applicable Technical Requirements and Standards.
- Installation of a breaker failure relay and scheme including DTT per standards outlined in PJM manual 7 and ComEd standards for the 345 kV gas circuit breaker.
- The Project Developer will install a SCADA system for interface with ComEd's SCADA system. Install 345 kV standard interconnection metering and telemetry to ComEd TSO including CB status, MW, MVAR, MWh and voltage values. Project Developer to co-ordinate updated metering/breaker status data exchange with ComEd SCADA Engineering.
- Power Output from the Project Developer shall be in accordance with the power quality standards contained in IEEE Standard 519. The generating units and all associated equipment at the Project Developer site shall not introduce any distortion of ComEd's waveform or telephone or carrier interference that is inconsistent or conflicts with such standard.
- Dual TRFM protection and site protection must be compliant with NERC & PJM requirements.
- Witness testing by ComEd or a DA will be required and must be pre-scheduled at least 90 days in advance.
- Project Developer to send a transfer trip to TSS 918 Dana for TR 345 kV CB breaker failure. Ability to isolate 87L and DTT functions is required. Example scheme/settings can be provided by ComEd.
- The Project Developer shall be responsible for installation and ownership of fiber between the

345 kV side of the solar farm and TSS 918 and the existing Wind Farm for breaker failure DTT with two physically diverse routes.

- At Osagrove Flats, it is assumed the new CTs from the 345kV MPT-2 breaker will be wired to the previously installed L91801 411L/311L relays. If this is deemed infeasible, ComEd and Project Developer design teams must discuss design alternatives at that point. Project Developer to modify relay settings as needed.
- For any new equipment connected to the BES (Bulk Electric System rated at 100kV or above) the associated primary/System 1 and secondary/System 2 protective schemes to have a minimum redundant:
    - Connected CTs (where available)
    - PT secondary (where available)
    - DC control circuits
    - Auxiliary trip relays
    - Circuit breaker trip coils (where available)
    - Communication circuitry
- ComEd shall supply and own any AMI meter devices. Project Developer shall be responsible for conveying access for any ComEd AMI metering equipment.
- A Project Developer with a proposed new Generating Facility that has a Maximum Facility Output equal to or greater than 100 MW shall install and maintain, at its expense, phasor measurement units (PMUs). See Section 8.5.3 of Appendix 2 to the Generator Interconnection Agreement as well as section 4.3 of PJM Manual 14D for additional information.
- The solar farm shall, at a minimum, be required to provide the Transmission Provider with site-specific meteorological data including:
    - Plane of Array (POA) Irradiance (Watts/meter^2)
    - Back Panel Temperature (Degrees Fahrenheit)
- All changes to topology, including generation, must be modeled during the Phase 1 study for PRC-027 compliance. A protection system coordination study is required for new BES buses or when there is a 15% (or greater) change in fault current for an existing BES bus. Setting changes may be required per the outcome of this coordination study.
- ComEd's Transmission Planning department will specify the HV tap setting for Osagrove Flats transformer MPT-2 at the time the transformer test reports are received.

B. Transmission Owner Facilities Study Results

The following is a description of the planned Transmission Owner facilities for the physical interconnection of the proposed AF2-143 project to ComEd transmission system. These facilities shall be designed according to ComEd Applicable Technical Requirements and Standards. Once built, ComEd will own, operate, and maintain these Facilities.

The proposed generation interconnection is shown in Attachment # 1.

1. TRANSMISSION OWNER INTERCONNECTION FACILITIES:

Utilize the 345 kV dead-end structure and foundation inside the fence of the Interconnection Substation, to terminate the Project Developer's generator lead line.

Utilize the line conductor from the dead-end structure to the bus position in the switchyard of the interconnection substation TSS 918 Dana.

This dead-end structure and foundation are included in AD2-038 (see assumptions in Section 6)

The proposed generation interconnection is shown in Attachment # 1.

2. STAND ALONE NETWORK UPGRADES

N/A

3. NETWORK UPGRADES

N/A

4. OTHER SCOPE OF WORK

- This facilities study report assumes the completion of installations specified in PJM New Service Request Project AD2-038.
- This facilities study report assumes the completion of PJM New Service Request Project AE1-163.
- ComEd to review and update existing TSS 918 DANA relay, meter, & SCADA settings for incorporation of new AF2-143 project.
- ComEd shall update CAPE model.
- The steady state voltage study for stability analysis will be performed by PJM during the Phase 2 System Impact Study Phase. The PJM study could identify upgrades to the ComEd system that would become part of this project's scope of work.
- ComEd shall Review and Update if required the AMI and Revenue Metering equipment. The review will take into consideration the new total generation output including auxiliary power requirements to be provided by Project Developer. The metering equipment will be sized to properly meter the output when the generation is online as well as the power supplied by ComEd for generator auxiliary power requirements when the generation is offline. All metering equipment shall be in accordance with ComEd Standards. Revenue grade metering must meet the requirements established by the published ComEd Interconnection Guidelines. Metering shall include registration of the generation output and auxiliary power usage.

5. MILESTONE SCHEDULE FOR COMPLETION OF [TO] WORK

Facilities outlined in this report are estimated to take 18 months to construct, from the time the Generation Interconnection Agreement is fully executed. This schedule may be impacted by the timeline for procurement and installation of long lead items, the ability to obtain outages to construct and test the proposed facilities.

Description	Start month	Finish month
Detailed Design	1	5
Permitting	-	-
Construction	5	18

6. ASSUMPTIONS IN DEVELOPING SCOPE/COST/SCHEDULE

- This facilities study report assumes the completion of installations specified in PJM New Service Request Project AD2-038.
- This facilities study report assumes the completion of PJM New Service Request Project AE1-163.
- This facilities study report assumes the new 345 kV circuit breakers shall be located within the same substation as L91801 Project Developer terminal and line relaying specified in AD2-038 Facility Study.
- ComEd estimate does not include costs of design and construction of TSS 918 Dana, AF2-143 Osagrove Flats Solar Farm, and transmission in Project Developer scope of work. ComEd estimated schedule is based on the GIA contract being executed by all parties.
- This cost estimates assume that work will be performed during normal weekdays and with no overtime. Transmission line outages for construction have not been identified, but generally are available from September to May. These outages are controlled by PJM.
- Costs are based on 2024 rates and do not reflect a potential increase in Labor or Material costs after 2024.
- ComEd cost estimate is valid for six (6) months after Facilities Study release by PJM.
- The Project Developer will be responsible to request and bear the cost for relocation of existing transmission or distribution lines (including structures and other facilities) that may be required for transmission line crossings, the transport of any large equipment, such as turbines, rotors, turbine structures, cranes, etc. Formal submittal of this request to ComEd's TSO for ultimate review by PJM can be made 7 months prior to back feed request date.
- Foundation design assumes typical soil conditions at locations and will be subject to change after soil boring tests.
- All upgrades to facilities included in this document will be required to meet latest ComEd standards.
- Upgrades are subject to change based on detailed design development.
- ComEd will complete pre-design and post construction survey for the transmission and substation upgrades, as required. This includes, but is not limited to, the LIDAR survey and video imaging for transmission lines. Costs associated with this are at the expense of the Project Developer. Pre-design survey must be completed prior to detailed engineering.
- This study assumes that there will be no additional right-of-way and/or easement work required.
- This Facilities Study is time dependent. If the project is not into construction within one year of the issuance, the study will be void and the project re-studied, requiring the completion of a new Facility Study.
- It is assumed that all associated network upgrades, as listed in the Phase 1 study, are complete prior to this New Service Request Project being placed in service.

7. REVENUE METERING REQUIREMENTS

All revenue metering needed for this interconnection project must meet the metering requirements stated in Appendix 2, section 8 of the AF2-143 GIA, and in PJM Manuals M01 and M14D. The details of applicable revenue metering requirements are given in the ComEd Interconnection Guidelines posted on PJM website.

The revenue metering will be installed on the ComEd side of the Point of Change in Ownership and will be installed, owned, and maintained by ComEd.

This revenue metering is included in AD2-038 (see assumptions in Section 6).

This Facility Study Report assumes that the initial interconnection, which is PJM New Service Request Project AD2-038, will include installation of metering equipment (including CT/PTs) that is designed to accurately meter the additional power output associated with this uprate or expansion. This assumption will be confirmed during engineering after a GIA is executed. Inability of the metering equipment to accurately meter necessary power flow may require additional metering installation.

- REVENUE METERING FOR PJM AND COMED
    - The revenue meter measures the wholesale energy output (Hourly compensated net MWH and Hourly compensated net MVARH) of the Generating Facility.
    - The metering equipment, including revenue meter and CT/PT shall be installed, at Project Developer's expense, at the interconnection substation on ComEd side of the Point of Change in Ownership.
    - ComEd shall own, operate, maintain, inspect, and test all the metering equipment as set forth in 'Testing of Metering Equipment' section of the PJM Tariff, at the Project Developer's expense.
- REAL-TIME METERING FOR PJM
    - The Project Developer shall install, own, operate, maintain, inspect, and test real-time metering equipment to measure and transmit directly to PJM the real time MW, MVAR, voltage and status of electrical equipment such as circuit breakers and Motor Operated Disconnect switches, in conformance with the requirements listed in PJM Manuals M-01 and M-14D, at the Project Developer's expense.
- RETAIL METERING FOR COMED
    - The AMI Meter measures the energy consumption by the Project Developer at transmission level and hence shall be designed to measure low MW flow.
    - The metering equipment including AMI Meter and CT/PT shall be installed at the interconnection substation on ComEd side of the POI, at the Project Developer's expense.
    - ComEd shall own, operate, maintain, inspect, and test all the metering equipment as set forth in the 'ComEd Interconnection Guidelines'.
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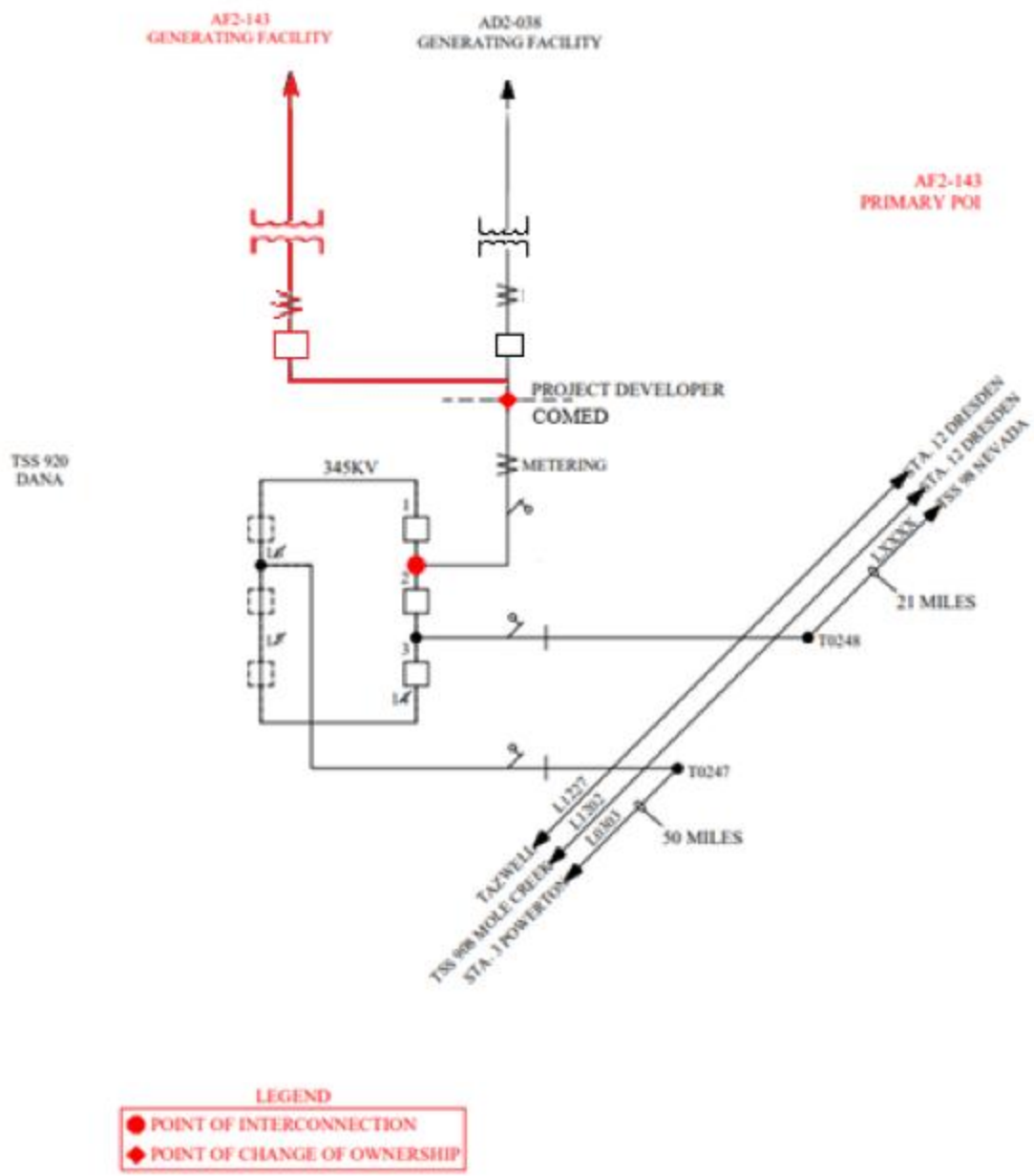
8. LAND REQUIREMENTS FOR INTERCONNECTION SUBSTATION

No additional easements, access rights, or temporary or permanent real property rights or acquisitions were identified as required for network upgrades to the ComEd system or for the project to interconnect at this location within this study. However, as further needs are assessed in detailed engineering, design and/or construction activities, if it is determined that there is a need for easements, access rights, or temporary or permanent real property rights or acquisitions, the developer is fully responsible for the costs to acquire these required land rights. Also, as necessary, the schedule will be adjusted accordingly to account for the necessary time to obtain these required land rights. All easements, access rights, or temporary or permanent real property rights or acquisitions shall comply with all ComEd requirements as detailed in “Land requirements for Interconnection Substations”.

9. ENVIRONMENTAL AND PERMITING

No environmental concerns and/or permitting requirements were identified as needed by this study. However, should detailed engineering and design and/or construction activities identify the need for an environmental study and/or permit requirements, the developer is fully responsible for the costs related to any environmental study, any actions to address the identified environmental impacts and the permits. Also, the schedule will be adjusted accordingly to account for the necessary time to perform the environmental study, address the identified environmental impacts and to obtain the permits, if applicable. All environmental studies, actions to address environmental impacts and permit actions shall comply with all ComEd requirements as detailed in "ComEd Environmental Requirements for Third Party Developers”, and with all local, city, county, state, and federal requirements.

C. APPENDICES

Attachment #1: Single line Diagram for the Physical Interconnection